Reflection

Min:

In this lab, our group analyze the Logistic Regression method that has been applied on the Titanic dataset for predicting whether passengers survived or not, using the confusion matrix heat map, classification report, and the 5-fold cross-validation method. The accuracy of the model reached about 80% for test data consisting of 179 passengers, however, the confusion matrix and the classification report demonstrated a better performance of the model in predicting passengers who died than those who survived. Practically speaking, the model could not identify 20 actual survivors, while wrongly predicting 15 people. This reveals how a single number of accuracies may conceal the mistakes made by a model. As a result of using the 5-fold cross-validation method for the entire dataset, the fold scores varied from 0.7753 to 0.8034, with the average score of 78.68% and the standard deviation of 0.0094. Since the accuracy score of the single split simulation was higher than all fold scores, it is most probably that it was lucky split and thus could be considered as the result of 5-fold cross-validation.

Chris:

Evaluating the performance of machine learning models in this lab helped me to learn more about the topic of model evaluation in addition to measuring the accuracy of a model. I learned how a confusion matrix maps out correct predictions and incorrect predictions. I learned about the measures of precision, recall, and the F1-score. Also, I learned more about cross-validation, i.e. testing a model on different splits of the data. The most important point is that the best measure for evaluation a model depends on the problem at hand. For a medical screening test, recall could be critical since a wrong negative result could have serious consequences for the patient. In this lab, I learned how to better evaluate my models and how to determine whether a model performs well enough on a problem.

Camillie:

Throughout this lab, I gained a better understanding of how machine learning models are evaluated beyond simply looking at accuracy. I learned the importance of precision, recall, and F1-scores and how each provides a different perspective on a model’s performance. Cross-validation also helped me understand why testing a model multiple times produces more reliable results than relying on a single test. One of my biggest

learning experiences was troubleshooting Python errors, which challenged me to pay closer attention to my code and understand how even small mistakes can affect the outcome. Overall, this lab strengthened my confidence in Python and helped me appreciate the importance of evaluating machine learning models for accuracy, consistency, and reliability.

Rigoberto:

My primary technical focus in this lab was implementing and evaluating our logistic regression model using K-fold cross-validation. Writing the `cross_val_score` script and executing a 5-fold evaluation cycle taught me how critical it is to move away from static data partitions. I observed firsthand how relying on a single, fixed train-test split introduces a massive element of "sampling luck," where a model's accuracy can be artificially inflated or deflated based purely on a random seed. Manually calculating the mean score and the standard deviation across all folds showed me how to audit a model's real generalizing power. Tracking standard deviation was a major breakthrough for me, as it proved that an algorithm's reliability relies just as much on low variance across different subsets of data as it does on a high overall accuracy score. On a technical level, troubleshooting a session state `NameError` by resetting our environment variables and forcing a linear, top-down script compilation reinforced the importance of sequence and asset caching in data pipelines. This exercise completely changed how I plan to validate machine learning frameworks moving forward.